

Machine Learning Model Evaluation Report

Wine Quality Classification using Random Forest

1. Model Overview

Property	Value
Model	Random Forest Classifier
Number of Trees	100
Training Samples	1279
Testing Samples	320
Number of Features	11

2. Model Performance

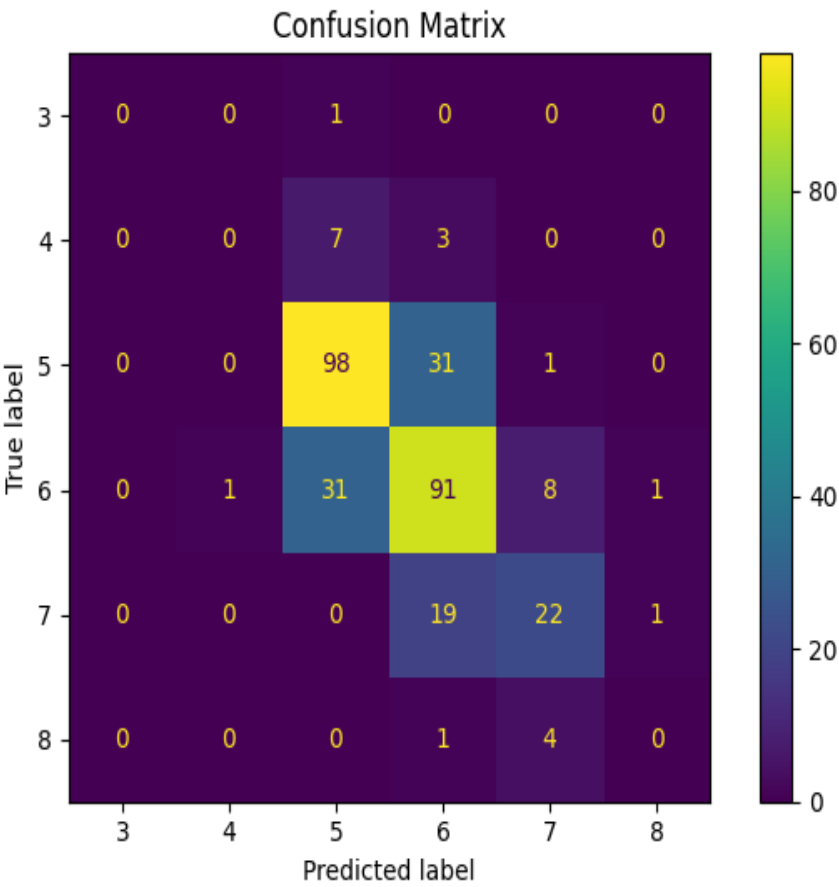
Metric	Score
Accuracy	65.94%
Precision	63.20%
Recall	65.94%
F1 Score	64.42%

3. Understanding the Metrics

Metric	Meaning
Accuracy	Percentage of all test predictions that were correct.
Precision	Among predictions made for a class, how often the prediction was correct.
Recall	Among samples that actually belong to a class, how many were correctly identified.
F1 Score	A balance between precision and recall. It is useful when both types of errors matter.
Confusion Matrix	Shows how actual classes were classified into predicted classes.

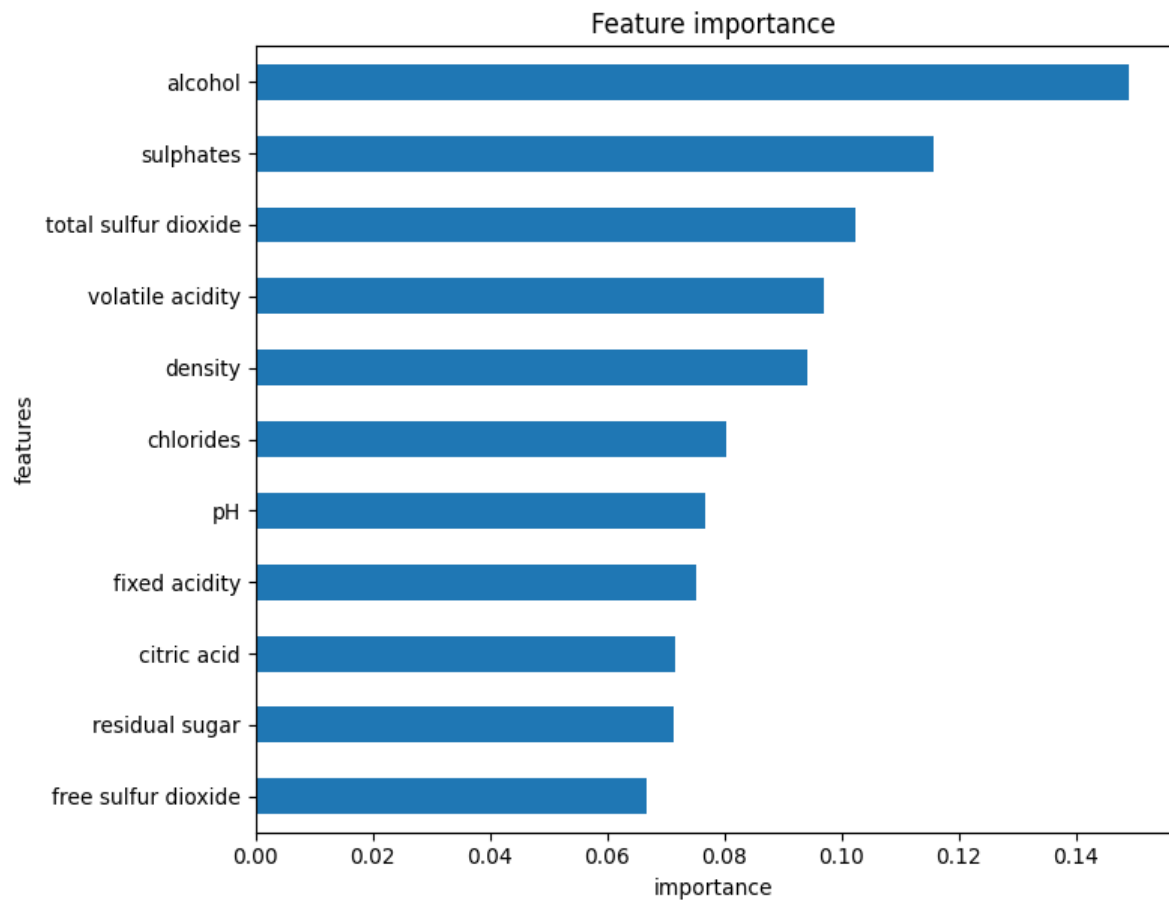
4. Confusion Matrix

The confusion matrix shows the relationship between the actual wine-quality classes and the classes predicted by the model.



5. Feature Importance

Feature importance indicates the relative contribution of features to the Random Forest's impurity-based splitting process.



6. Model Findings

The Random Forest model achieved an accuracy of 65.94% and a weighted F1 score of 64.42%. The model achieved moderate overall predictive performance.

The model performs better on the more frequently represented quality classes and has difficulty identifying rare classes. This indicates that class distribution is an important consideration when interpreting the results.

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